Correlation and Regression

Introduction

In data science (and by extension, AI), we are often concerned with the relationships between two or more variables (or features) of a dataset.

• Each data point in the dataset is an observation, and the features are the properties or attributes of those observations.

• Example question – What mathematical function exists between population density and the GDP per capita of different countries?

Lesson 2 – Correlation

- Correlation aims to find the degree of relationship between two variables, x and y.
- A **scatter plot** is the best method to visually represent the relationship between two variables.
- A correlation could be:
    - **Positive**: both variables change in the same direction.
    - **Negative**: variables change in opposite directions.
    - **Neutral**: No relationship in the change of the variables.

Lesson 3 – Correlation vs Causation

Correlation does not mean causation, although they might exist at the same time.

If action/variable *x* and outcome *y* are correlated that does not necessarily mean *x* caused *y*, because...

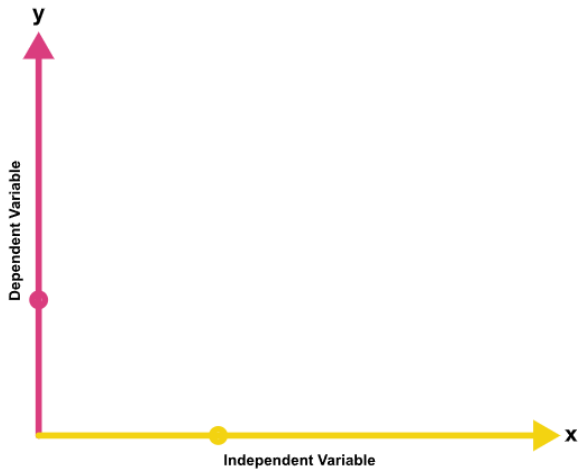
Outcome *y* could be caused by different action/variable ***z***	Outcome ***y*** could be caused by ***x*** but only when ***v*** happens	There is a chain of causes within ***x*** *and* ***y***	Or it could actually be the other way around – ***x*** is caused by ***y***

Lesson 4 – Quantifying Correlation

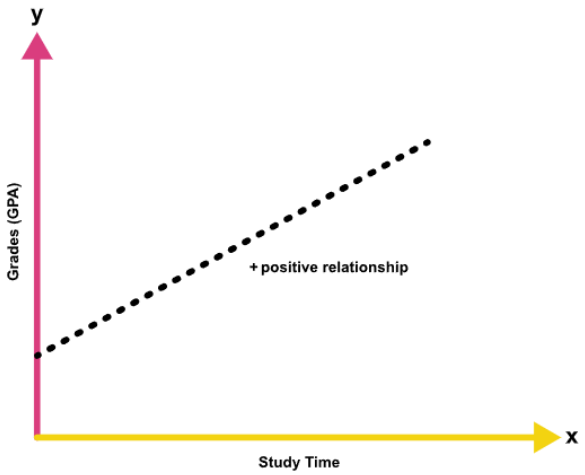
Covariance	Pearson Correlation Coefficient

Both of these techniques only measure linear relationships.

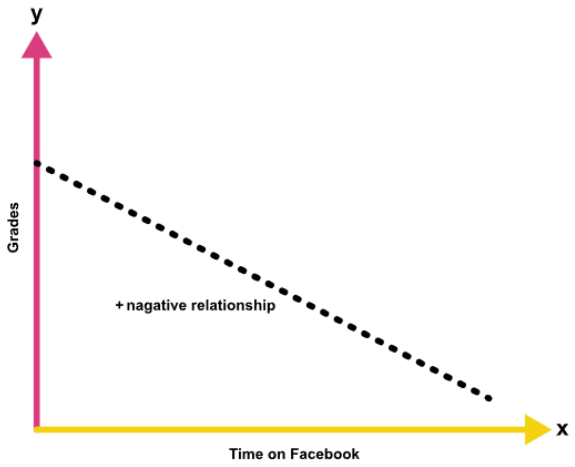
Lesson 5 – Independent & Dependent Variables



This first graph shows the position of the dependent and independent variables.



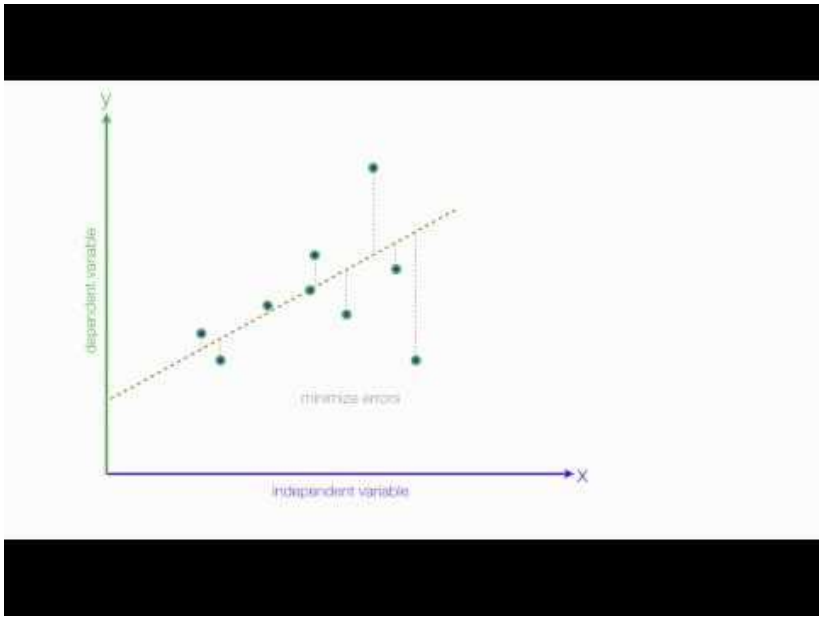
This graph shows a positive relationship.



This graph shows a negative relationship.

Lesson 6 – Regression

For a recap on regression, see the Numerical Analysis module. You can also review these videos on the subject, including a worked example.



Lesson 7 – Measuring Covariance

Measuring Covariance is the measure of how two random variables (x and y) change together.

Each individual x value is subtracted from the mean of x and similarly for y. These are

multiplied together and the sum of all the outcomes is taken. The sum is divided by the total

number in the sample. This final value represents the covariance between two variables.

Interpreting Covariance Values

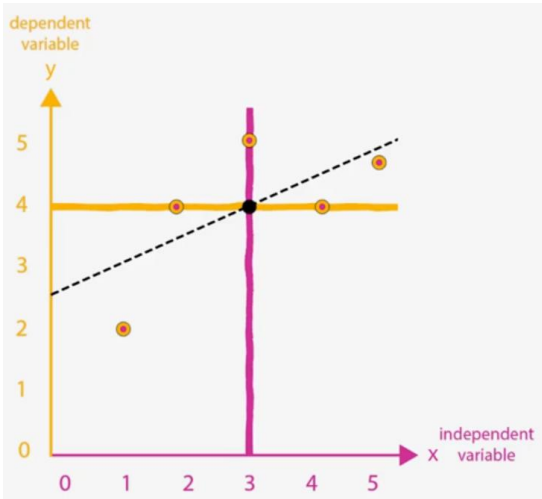
Sign of covariance

(+) → two variables are moving in same direction

(-) → two variables are moving in opposite directions.

Size of covariance: if the number is large, the strength of correlation is strong

This example shows how to calculate the covariance from the regression analysis. What can you say about the covariance in this example?



	x	y	x - x̄	y - ȳ	(x - x̄)²	(x - x̄)(y - ȳ)
	1	2	-2	-2	4	4
	2	4	-1	0	1	0
	3	5	0	1	0	0
	4	4	1	0	1	0
	5	5	2	1	4	2
mean	3	4			10	6

$$cov(x,y) = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{n}$$

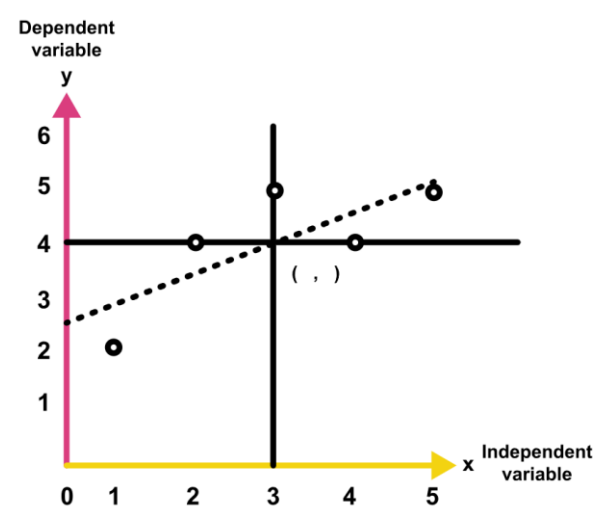
$$cov(x,y) = \frac{6}{5} = 1.2$$

Lesson 8 – Interpreting Covariance Values

Sign of Covariance:

(+) two variables are moving in same direction.	(+) two variables are moving in same direction.

Size of Covariance: If the number is large, the strength of correlation is strong.



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Mean	3	4				
Sum					10	6

$$cov(x, y) = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{n}$$

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This example shows how to calculate the covariance from the regression analysis. What can you say about the covariance in this example?

Lesson 9 – Limitation

The covariance is dependent on the variability in the data. So a large variance gives large numbers. Therefore, the magnitude cannot be measured, that is why it is difficult to interpret.

So, what is the solution??

Lesson 10 – Limitations of Pearson's r

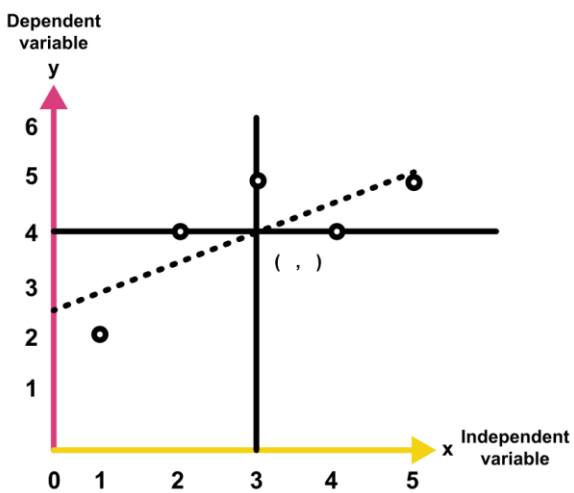
It is sensitive to outliers, that is, an outlier can affect the *r* value.	It cannot be used to predict anything but a linear relationship, ignoring many other types of relationships or correlation.

Lesson 11 – Linear Regression

Correlation is the premise for regression. Once a correlation between the independent and dependent variable is established, can a dependent variable be predicted when the independent variable is changed?

Lesson 12 – Assumption

1. Linear relationship
2. Observations are independent
3. Residuals (difference between predicted and observed values of dependent variables) are normally distributed
4. Residuals have the same variance



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Mean	3	4				
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- *Y* hat (predicted value of the dependent variable)
- b_0 = y intercept (where the regression line intercept y axis)
- b_1 = slope of the line
- *x* = value of the independent variable

Now solve what would be the predicted value of dependent variable if the independent variable value is 24?

Predicted dependent variable = *y* intercept + slope ∗ independent variable

$$\hat{Y} = b_0 + b_1 x$$

$$b_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2} = \frac{6}{10} = 0.6$$

Solving b_0 with the mean values (3,4)

$4 = b_0 + 0.6 * 3$

$b_0 = 2.2$

Lesson 13 – Multiple Linear Regression

Multiple linear regression is used to predict the correlation between more than one independent variables and a dependent variable. e.g., car engine volume and weight is correlated with CO2 emission.

When the data is plotted on a graph, there appears to be a hyperplane relationship.

Simple Linear Regression involves one independent variable and is shown below.

Multiple Linear Regression involves multiple independent variables and the equation is given below.

You will be able to work through some examples in the activities of this Unit (see the learning platform).

Lesson 14 – Logistic Regression

Regression can also be used for classification. A type of regression analysis called logistic regression can predict a binary outcome (yes/no) based on a set of independent variables. Due to the binary nature of the classification logistic regression is known as a binary classification.

For definitive multi-class classification, we use decision tree type classifying algorithms. These video tutorials clearly describe Logistic Regression and Decision Trees with applicable examples and codes and using datasets from Kaggle:

References

Holmes, A., Illowsky, B., Dean, S. (2017) Introductory Business Statistics. Houston, Texas. Available from https://openstax.org/books/introductory-business-statistics under a Creative Commons Attribution 4.0 International license (CC BY 4.0)

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